The Maximum Number of Intersection Points of Overlapping Regular Polygons

With convexity proofs and a region count

Can Ertürk
An award-winning project at MOSTRATEC (Mostra Internacional de Ciência e Tecnologia)
Halil İnalcık Bilim ve Sanat Merkezi (BİLSEM), Bursa, Türkiye
Revised and extended, 2026. Author currently a BSc student at Eindhoven University of Technology (TU/e).

Abstract. This note asks how many intersection points can appear at most when several regular polygons are laid over one another in the plane. Because a regular polygon is convex, two regular k-gons can cross in at most 2k points, and the proof of this bound is short. For n copies of the same k-gon the count grows to $kn^2 - kn$, and similar closed forms hold for mixtures of polygons of different sizes. The general rule given in the original project is restated here with its index error corrected. The note also settles a question left open earlier: the same arrangement cuts the plane into exactly $kn^2 - kn + 1$ bounded regions. Each formula was checked by a short computer program over many cases.

Keywords: *combinatorial geometry, convex polygons, arrangements, intersection points, Euler's formula.*

1. Introduction

Suppose a few regular polygons are drawn on top of each other in the plane. Their edges cross, and the question of this work is simple to state: at most how many crossing points can there be? Counting problems of this kind sit at the start of combinatorial geometry, and the answers turn out to be useful in tiling, in structural design, and in geometry-based methods for hiding information.

The guiding idea is that the largest count occurs when the polygons are in general position: every pair of them crosses, and no point is shared by three or more polygons at once. When that holds, each pair contributes its own maximum and none of the points coincide, so the totals just add together. The rest of the note makes this precise and proves it.

Compared with the original project, three things are new here. First, the bound for a single pair is proved rather than only observed. Second, the general rule for mixed polygons is corrected. Third, the number of regions the arrangement creates is worked out.

2. How the results were found and checked

The shapes were first drawn in GeoGebra. A second polygon was placed at the same centre and turned by a fraction of the base polygon's symmetry angle, just enough that the edges crossed cleanly without any accidental overlaps. The crossing points were counted by hand, tabulated, and the pattern across triangle, square and pentagon was generalised by algebra.

For this revision every formula was also tested by a short program that finds all crossings between polygon edges and discards duplicates. It was run on every combination of k in {3,4,5,6} and n in {2,3,4,5}, and the region count was checked the same way with a face count. The program agreed with the formulas in all cases. The code is available with this note.

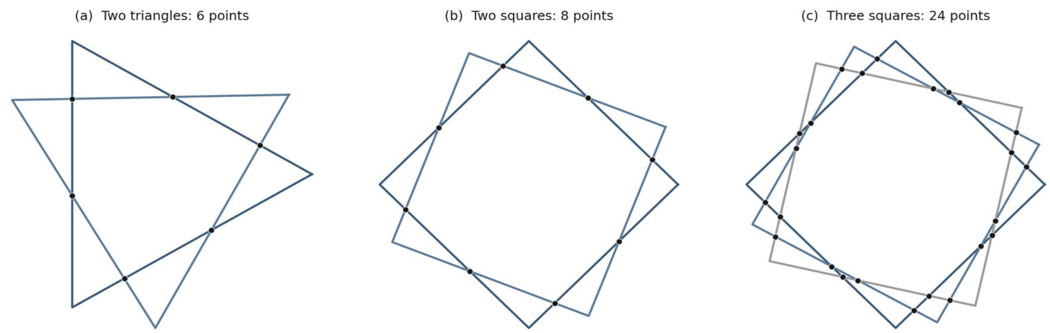

Figure 1. Polygons laid over one another, with the crossing points marked. From left: two triangles give 6 points, two squares give 8, three squares give 24.

3. Two polygons

Theorem 1. *Two regular k-gons cross in at most 2k points, and 2k can be reached.*

Proof. A regular polygon is convex, so its outline is a convex closed curve. A straight line can enter and leave such a curve only once, so it meets the outline in at most two points. Each of the k edges of the first polygon is a straight segment, so each meets the second polygon's outline at most twice. That gives at most 2k crossings in total. Turning one polygon slightly about the shared centre produces a picture in which every edge does cross twice (Figure 2), so the value 2k is actually attained. ■

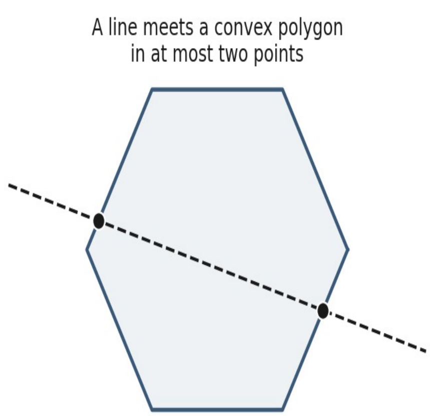

Figure 2. The reason behind the bound: a line meets a convex polygon in two points at most.

A note on the limit. For polygons that are allowed to be non-convex, two simple polygons with m and n sides can cross many more times, up to mn when both side counts are even. Those configurations need shapes that bend back on themselves. A regular polygon never does, so the convex limit 2k is the one that applies here, and Theorem 1 reaches it.

4. Many copies of one polygon

Theorem 2. *For n regular k-gons in general position, the largest number of crossing points is* $kn^2 - kn$*, equal to* $kn(n-1)$*.*

Proof. There are $C(n,2) = n(n-1)/2$ pairs of polygons. By Theorem 1 each pair gives at most 2k points, and in general position no two of these points fall on top of each other. Multiplying, the total is $n(n-1)/2$ times 2k, which is $kn(n-1)$, or $kn^2 - kn$. ■

Putting in the value of k recovers the rules from the original project.

Polygon	Sides k	Most points for n copies
Triangle	3	$3n^2 - 3n$
Square	4	$4n^2 - 4n$
Pentagon	5	$5n^2 - 5n$
Regular k-gon	k	$kn^2 - kn$

Table 1. The rule for n equal polygons and its special cases.

5. Mixing polygons of different sizes

The same pairwise reasoning carries over to polygons of different sizes. Two convex polygons with k_1 and k_2 sides cross in at most $2\min(k_1, k_2)$ points, since each edge still meets the other outline at most twice and the smaller side count is what limits the total. Two cases come up often, written with the smaller polygon having k sides and k less than n:

$$\text{two } n\text{-gons and one } k\text{-gon:}\quad 2n + 4k \qquad\qquad \text{three } n\text{-gons and one } k\text{-gon:}\quad 6(n + k)$$

Take three n-gons and one k-gon. The three n-gon pairs give 3 times 2n, which is 6n, and the three n-with-k pairs give 3 times 2k, which is 6k. Adding these is $6(n + k)$. The other case follows the same way, and the program confirms both.

5.1 The general rule, corrected

Take a_1 polygons with k_1 sides, a_2 with k_2 sides, and so on, with $k_1 < k_2 < \ldots < k_m$. The largest number of crossings is T_1 plus T_2. Here T_1 counts crossings between polygons of the *same* size, and T_2 counts crossings between polygons of *different* sizes:

$$T_1 = \Sigma_i\, k_i\, (a_i^2 - a_i)$$

$$T_2 = \Sigma \text{ over } i < j \text{ of } \; 2\min(k_i, k_j)\, a_i\, a_j \;=\; \Sigma \text{ over } i < j \text{ of } \; 2\, k_i\, a_i\, a_j$$

What was fixed. The original write-up of T_2 mixed up its indices, with terms like $(a_1^2 - a_3)$ that have no clear meaning. The corrected T_2 above simply runs once over each pair of polygon types of sizes $k_i \le k_j$ and adds $2 \min(k_i, k_j)\, a_i\, a_j$, which is $2 k_i a_i a_j$. The full answer is $T_1 + T_2$.

6. Counting the regions

The original project listed, as a question for later, how many *regions* the crossings carve the figure into. That question has a clean answer.

Theorem 3. *For n regular k-gons in general position, the overlaid edges enclose exactly $kn^2 - kn + 1$ bounded regions, which is $P + 1$ where $P = kn^2 - kn$.*

Proof. Read the union of the outlines as a graph drawn in the plane. Its corner points are the kn original polygon vertices together with the P crossing points, so $V = kn + P$. Each crossing splits two edges into two, which adds two segments, so $E = kn + 2P$. For a connected drawing like this, Euler's formula gives the number of enclosed regions as $E - V + 1$, which works out to $(kn + 2P) - (kn + P) + 1$, that is $P + 1$. ■

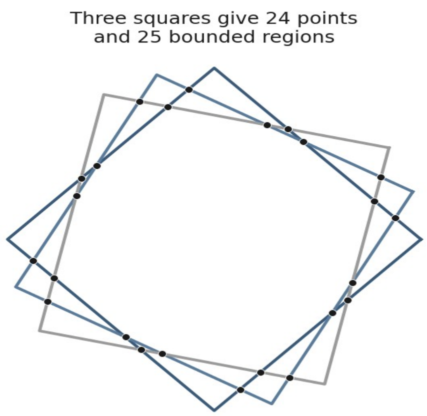

Figure 3. Three squares: 24 crossing points and 24 + 1 = 25 enclosed regions, as Theorem 3 predicts.

A direct region count by the program matched $P + 1$ in every case tried. This region formula is the main new result of the revision.

7. Summary

(1) Two regular k-gons cross in at most 2k points; the bound follows from convexity.

(2) n equal k-gons in general position give $kn^2 - kn$ points: triangles $3n^2 - 3n$, squares $4n^2 - 4n$, pentagons $5n^2 - 5n$.

(3) Two n-gons and one k-gon give $2n + 4k$; three n-gons and one k-gon give $6(n + k)$, for k less than n.

(4) General mixture: $T_1 + T_2$, with $T_1 = \Sigma\, k_i(a_i^2 - a_i)$ and $T_2 = \Sigma$ over $i < j$ of $2 \min(k_i, k_j)\, a_i\, a_j$.

(5) The enclosed regions number $kn^2 - kn + 1$, one more than the crossing points (Theorem 3).

8. Where this could go next

- Polygons that are not convex, such as stars, where the limit $2k$ no longer holds and the larger mn count becomes possible.
- The same questions in three dimensions, for overlapping regular solids, counting points, edges and regions.
- Colouring the regions or measuring their areas, building on the region count of Section 6.

9. Where the bounds may be useful

The counts proved here are small and exact, which is what makes them handy as bounds in problems where overlapping convex shapes appear. Three settings stand out. In each, the link is deliberately stated as a bound or a special case, not as a finished tool.

9.1 Chip layout and lithography

Integrated-circuit layouts are built from overlapping polygons on stacked mask layers, and a recurring task in design-rule checking is to work out where the layers cross and which closed regions they form. Real layout shapes are rectilinear rather than regular, so the formulas here do not apply directly. What they do give is a clean, proved instance of the underlying question: for the regular convex case, the number of crossings between layers is at most $2k$ per pair, and the closed regions number exactly $P + 1$. This is the kind of exact bound that region-extraction and boolean-layer routines try to respect, and the regular case is a natural test case for them.

9.2 Convex shapes in motion planning

In robot motion planning, vehicles and obstacles are often modelled as convex polygons, and collision tests reduce to asking whether two convex outlines meet. The result of Section 3 bounds the size of that intersection: two convex k-gons share at most $2k$ boundary points, so any routine that lists the crossing points has at most $2k$ of them to handle per pair. The contribution here is not a faster collision test, since good ones already exist, but a clean ceiling on the output size of the intersection step, which is useful when reasoning about worst-case cost.

9.3 Cellular structures in generative design

Lightweight structures such as honeycomb and lattice panels are arrangements of repeated convex cells, and design software increasingly searches for the layout that is strongest for its weight. A closed form for the number of vertices and enclosed regions of such an arrangement gives that search a cheap geometric quantity to use as a constraint or as part of a score, instead of recomputing the figure each time. The formulas here supply exactly that quantity for the regular case.

A note on scale. In geographic data, driving maps, or large CAD models, millions of polygons may overlap at once. Counting their crossings at that scale is mainly an engineering problem of parallel computation rather than a question about the bounds proved here, and it sits outside the scope of this note.

References

[1] M. B. Dillencourt, D. M. Mount, A. Saalfeld. On the maximum number of intersections of two simple polygons, 1993.

[2] B. Poonen, M. Rubinstein. The number of intersection points made by the diagonals of a regular polygon. SIAM J. Discrete Math., 11(1):135–156, 1998.

[3] An almost optimal bound on the number of intersections of two simple polygons. Discrete & Computational Geometry, 2022.

The figures, table and computer checks in this note were produced by the author using GeoGebra and a Python script.